



Image Deconvolution and Point-spread Function Reconstruction with STARRED: A Wavelet-based Two-channel Method Optimized for Light-curve Extraction

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Abstract

We present STARRED, a point-spread function (PSF) reconstruction, two-channel deconvolution, and light-curve extraction method designed for high-precision photometric measurements in imaging time series. An improved resolution of the data is targeted rather than an infinite one, thereby minimizing deconvolution artifacts. In addition, STARRED performs a joint deconvolution of all available data, accounting for epoch-to-epoch variations of the PSF and decomposing the resulting deconvolved image into a point source and an extended source channel. The output is a high-signal-to-noise-ratio, high-resolution frame combining all data and the photometry of all point sources in the field of view as a function of time. Of note, STARRED also provides exquisite PSF models for each data frame. We showcase three applications of STARRED in the context of the imminent LSST survey and of JWST imaging: (i) the extraction of supernovae light curves and the scene representation of their host galaxy; (ii) the extraction of lensed quasar light curves for time-delay cosmography; and (iii) the measurement of the spectral energy distribution of globular clusters in the “Sparkler,” a galaxy at redshift $z = 1.378$ strongly lensed by the galaxy cluster SMACS J0723.3-7327. STARRED is implemented in JAX, leveraging automatic differentiation and graphics processing unit acceleration. This enables the rapid processing of large time-domain data sets, positioning the method as a powerful tool for extracting light curves from the multitude of lensed or unlensed variable and transient objects in the Rubin-LSST data, even when blended with intervening objects.

Unified Astronomy Thesaurus concepts: [Deconvolution \(1910\)](#); [Photometry \(1234\)](#); [Light curves \(918\)](#); [Astronomy image processing \(2306\)](#)

1. Introduction

The resolution of astronomical images is fundamentally limited by the atmospheric turbulence in the Earth’s atmosphere or by optical diffraction of the telescope in the case of space observations. Both of these effects induce blurring of the images, which can be corrected by knowing the point-spread function (PSF), a linear mapping between the flux intensity in the sky and that actually recorded at the focal plane of the instrument. However, the PSF evolves temporally, with changing atmospheric conditions.

Temporal variations of the PSF reduce the photometric accuracy that can be achieved for blended sources, since light from adjacent sources is mixed differently in changing seeing conditions. This introduces systematic errors in the light curves that, unlike photon noise, cannot be reduced with longer exposures (systematics in fact actually get worse with an increasing signal-to-noise ratio or SNR). This typically affects lensed quasars, as the light of the lensed images is often blended with the foreground lensing galaxy shear measurement, as recently demonstrated in Finner et al. (2023).

The epoch-to-epoch variations of the PSF can be corrected by modeling the PSF of each individual image and then removing its effect from the data, a problem known as PSF deconvolution. This inverse problem is mathematically ill-

posed, due to pixelization and the presence of noise resulting in the absence of a unique and stable solution. To address this, one can introduce regularization to the solution, essentially acting as a Bayesian prior that constrains the parameter space for potential solutions. Regularization selects the best solution among the plethora of possible ones, given some prior information embedded in the regularization term.

Historically, deconvolution techniques such as that of Tikhonov (Tikhonov & Arsenin 1977), characterized by a quadratic regularization term and a closed-form solution, and its related Wiener deconvolution (Sekko et al. 1999) have been pivotal. Others have postulated that the optimal solution must be the one that has maximum entropy (Skilling & Bryan 1984; Narayan & Nityananda 1986). CLEAN (Högbom 1974; Cornwell 2008), a well-known early method to image deconvolution for sky brightness estimation, was mostly employed in radio astronomy. With the arrival of the Hubble Space Telescope (HST), the Richardson–Lucy algorithm (Richardson 1972; Lucy 1974) became popular. It implements an iterative process based on a Bayesian approach to estimate the most likely model from a noisy image, accounting for the influence of the PSF. This algorithm was later extended to a two-channel decomposition in Pirzkal et al. (2000), Becker et al. (2003), and Velusamy et al. (2008), specifically designed for separating point sources and extended sources. Subsequent advancements involved introducing prior assumptions in the form of a hierarchical parameter model (Selig & Enßlin 2015).

The advent of deep neural networks (DNNs) has improved performance and inference times by leveraging existing



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databases and learning features in a supervised fashion (Sureau et al. 2020; Akhaury et al. 2022; Nammour et al. 2022; Adam et al. 2023). To date, all of these methods have attempted to correct for the full PSF, resulting in the Nyquist–Shannon sampling theorem not being satisfied (Magain et al. 1998): a deconvolved point source, which is essentially a Dirac function, can never be accurately represented by any sampling or pixel size. Deconvolving by the full PSF is therefore prone to producing unwanted image artifacts (“ringing”), since the deconvolved image is to be represented on an array of pixels, regardless of its size. In general, these artifacts can be suppressed by the inclusion of prior information, either in the form of a regularization term in a likelihood-based approach or via the training sets for deep-learning-based methods. Typically, a DNN will require a very large training set of images that have better resolution than the data, which is challenging to generate. For these reasons, both deep learning and standard approaches should be considered complementary.

Addressing this challenge of partial deconvolution, Firedec (Cantale et al. 2016) proposes a two-channel method to improve the resolution of the data rather than completely eliminating the effect of the PSF. Specifically, it decomposes the images into a parametric point-source channel and an extended source channel, represented on a pixel grid and enforcing wavelet regularization. However, the Haar wavelets considered in Firedec are not shift-invariant and are not well adapted to the typical topology of astronomical objects.

Wavelet regularization introduces a minimally informative prior to penalize the presence of nonzero elements in the wavelet domain. An isotropic wavelet basis in which astronomical objects are well represented, i.e., *sparsely* represented, is starlets (Starck et al. 2015), meaning that they can be expressed as a linear combination of a few atoms from the starlet dictionary. A sparse representation of the signal is a desired property as it facilitates image denoising. Since the noise is not sparse in the starlet domain, it can be reduced by reconstructing the image only from the most significant coefficients, in a technique called *image thresholding*. The different scales resulting from the starlet transform can be penalized differently, with the noise being mostly represented on the finer wavelet scale.

In this paper, we present the mathematical framework as well as several tests and science applications of STARRED, a method we previously introduced in Michalewicz et al. (2023). STARRED performs both the PSF reconstruction and the image deconvolution steps, based on the principles of MCS (Magain et al. 1998) and Firedec (Cantale et al. 2016), but with various key enhancements. These include the use of a family of isotropic wavelets (starlets) specifically designed to represent astronomical objects, automatic differentiation, accelerated execution, scalability on graphics processing units (GPUs), and the capability to handle multi-epoch deconvolution.

The method is very general and has a wide range of applications. These include but are not limited to light-curve extraction of variable objects such as active galactic nuclei or supernovae, point-source photometry in crowded fields, or the quasar/host decomposition problem (see, e.g., Leist et al. 2024 for a recent comparison between existing methods). Here, we select three science cases to demonstrate the capabilities of STARRED: (i) light-curve extraction and scene modeling of type Ia supernovae (SNe Ia); (ii) time-delay measurement in lensed quasars; and (iii) spectral energy distribution (SED) measurement of high-redshift globular clusters. We apply this method to

simulated Rubin-LSST time series and to real JWST observations, assessing the performance of the algorithm in terms of photometric and astrometric accuracy and computational speed.

The paper is organized as follows. In Section 2, we present the mathematical formalism of the method. In Sections 3 and 4, we assess the performance of the PSF reconstruction and image deconvolution methods, respectively.⁶ We discuss our results in Section 5 and conclude in Section 6, also mentioning the limitations and future improvements of STARRED.

2. Formalism

The maximum resolution that a deconvolution algorithm can achieve is fundamentally limited by the Nyquist–Shannon sampling theorem, which states that for an evenly sampled signal, and in the absence of any additional information, no frequencies higher than twice the sampling frequency can be recovered. Historically, deconvolution algorithms ignored this theorem, by deconvolving with the observed PSF, hence attempting to recover an infinitely high resolution. The result of this deliberate choice of going beyond the Nyquist–Shannon limit is illustrated in one dimension in Figure 1. The deconvolved image is free of artifacts only if the signal is perfectly centered on the pixel grid, which is the only configuration allowing the sampling of a Dirac function. As soon as the signal is (even slightly) off-centered with respect to a pixel, aliasing immediately appears in the deconvolved output. However, if a narrower PSF is chosen for deconvolution, as illustrated in the last row of Figure 1, the resulting output satisfies the Nyquist–Shannon limit and the signal is reconstructed without artifacts. This remains valid in two dimensions when considering image deconvolution.

2.1. PSF Reconstruction

To circumvent the issue mentioned above, STARRED follows MCS and Firedec in not attempting to completely remove the PSF effect when reconstructing images on pixel grids. It instead proposes to decide in advance the resolution and even the shape of the PSF in the deconvolved image, ensuring that it is well sampled given the pixel size adopted to represent the deconvolved image. Of course, this pixel size can well be smaller than in the original data, so that the gain in resolution can be very effective even if the sampling of the original data is not good (e.g., HST or JWST images). The choice made in STARRED is that the PSF in the deconvolved image is an isotropic Gaussian with two pixels at FWHM. This choice, which adheres to the Nyquist–Shannon sampling theorem, involves reconstructing a so-called *narrow* PSF s , which relates to the original PSF t through

$$t = (s * r)^\downarrow, \quad (1)$$

where r is a Gaussian with an FWHM of 2 pixels and $\downarrow$ performs the subsampling operation (Magain et al. 1998). We decompose s as a sum of an analytical Moffat function $m(\eta_{\text{Moffat}})$ and a grid of pixels $b(\eta_{\text{pixel}})$, such that

$$s(\eta_{\text{PSF}}) = m(\eta_{\text{Moffat}}) + b(\eta_{\text{pixel}}), \quad (2)$$

where $\eta_{\text{PSF}} = (\eta_{\text{Moffat}}, \eta_{\text{pixel}})$ are the free parameters of the model. This narrow PSF is the one used in the rest of the work,

⁶ Examples and tests presented in this paper can be reproduced from this Gitlab repository: <https://gitlab.com/cosmograil/starred-examples>. A copy of these materials has been deposited to Zenodo: [doi:10.5281/zenodo.11212469](https://doi.org/10.5281/zenodo.11212469).

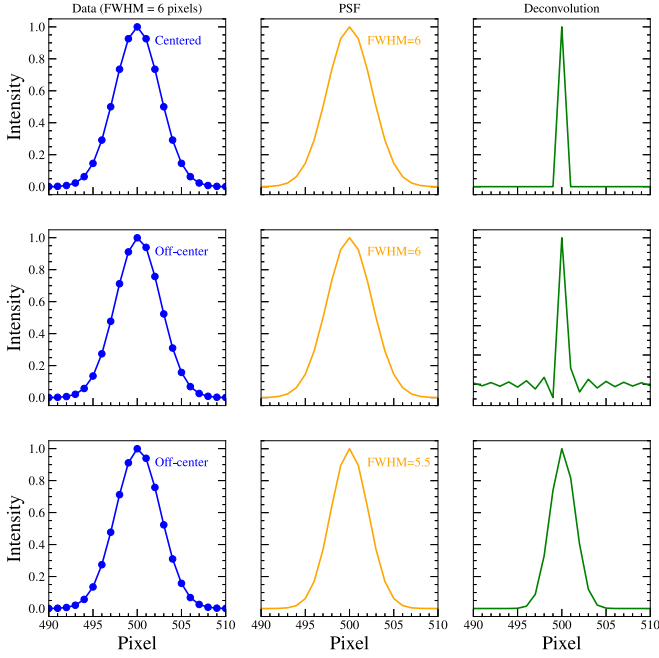


Figure 1. Illustration of Gibbs oscillations due to the aliasing of high frequencies in the frequency domain allowed by the pixel size. Top: from left to right, a noise-free point source is perfectly centered on a pixel and is deconvolved with a PSF that has the same FWHM. Here, we use direct Fourier division to do the deconvolution, as there is no noise involved. The deconvolved point source is a delta function also perfectly centered on a pixel. All frequencies, even infinite, are well represented regardless of the pixel size, i.e., no artifacts appear. Middle: the exact same experiment, but now the signal is off-centered by 0.1 pixels with respect to the center of a pixel. High frequencies are not well represented and Gibbs oscillations do appear. Note also that (i) the mean flux level is now wrong; and (ii) the impact of the Gibbs oscillations depends on where a given star in the data falls with respect to the pixels, with the worst effect corresponding to stars appearing exactly between two pixels. Bottom: the same as in the middle row, but now the PSF used to deconvolve is narrower than the data, hence the deconvolved image does not contain infinitely high frequencies that are impossible to sample. The deconvolved point source has a finite resolution and does not present Gibbs oscillations.

in place of the observed PSF, and can be constructed from stars in the field of view. In Figure 1, t corresponds to the left column, s corresponds to the middle column, and r is the right column. In other words, s is the kernel that transforms t into r , where the latter is an isotropic Gaussian profile of any chosen FWHM. From a star cutout t_k containing noise n , one can solve the following inverse problem to obtain an estimate of s :

$$t_k = (s * a_k r_k)^\dagger + n, \quad (3)$$

where r_k is a shifted Gaussian such that $r_k(x) = r(x - c_k)$, with c_k being the center of the Gaussian and a_k its amplitude. By defining B_k as the (shifted) Gaussian convolution operator for the k th star and D as the downsampling operator, we write the data fidelity term as the generalized χ^2 of the model for a set of N stars:

$$\mathcal{L}(t | \eta_{\text{PSF}}, a_k, c_k) = \frac{1}{2} \sum_{k=1}^N [t_k - D B_k(a_k, c_k) s(\eta_{\text{PSF}})]^\top C^{-1} [t_k - D B_k(a_k, c_k) s(\eta_{\text{PSF}})], \quad (4)$$

where C^{-1} is the inverse of the noise covariance matrix. Here, we assume spatially uncorrelated noise, so C is a diagonal matrix. Of course, nothing prevents us from applying STARRED

to data with correlated noise, by using the full noise covariance matrix. This problem is unstable in the presence of noise in the data, which requires adding some prior information to the model parameters η_{pixel} to constrain the solution. The loss function L thus reads

$$L(\eta_{\text{PSF}}, a_k, c_k) = \mathcal{L}(t | \eta_{\text{PSF}}, a_k, c_k) + \mathcal{P}(\eta_{\text{pixel}}), \quad (5)$$

the log-likelihood first term capturing the data fidelity and the second being a prior or regularization term. The latter is acting in our case only on a subset of the parameters η_{pixel} and is defined as:

$$\mathcal{P}(\eta_{\text{pixel}}) = \lambda \| W \odot \Phi^\top b(\eta_{\text{pixel}}) \|_1, \quad (6)$$

where $\|\cdot\|_1$ corresponds to the ℓ_1 -norm, λ is the regularization strength, Φ is the starlet transform operator, and $\odot$ is the Hadamard product, i.e., the element-wise product. This prior is equivalent to applying a soft-thresholding of the starlet coefficients with a level based on the hyperparameter λ , expressed in noise standard deviation units (Starck 2016). The matrix W weights each starlet coefficient by its corresponding noise (see Section 2.3 and Joseph et al. 2019 for more details). λ is crucial hyperparameter to tune the strength of the regularization. It is often set between 1 and 3 to ensure the high significance of the reconstructed signal. Setting $\lambda=3$ means that only the coefficients exceeding the noise level by 3σ will be reconstructed.

Overall, the minimization problem under the above sparsity constraint reads

$$\arg \min_{\eta_{\text{PSF}}, a_k, c_k} \frac{1}{2} \sum_{k=1}^N [t_k - D B_k(a_k, c_k) s(\eta_{\text{PSF}})]^\top C^{-1} \times [t_k - D B_k(a_k, c_k) s(\eta_{\text{PSF}})] + \lambda \| W \odot \Phi^\top b(\eta_{\text{pixel}}) \|_1. \quad (7)$$

2.2. Two-channel Multi-epoch Deconvolution

Given the *narrow* PSF at the i th epoch, s_i , the single-epoch deconvolution problem can be formulated as follows:

$$d_i = (s_i * f_i)^\dagger + n_i, \quad (8)$$

where d_i is the original image, f_i is the deconvolved higher-resolution image, and n_i is the noise. In STARRED, f_i is modeled as the sum of the contributions of two distinct *channels*, a point-source channel and an extended source channel:

$$f_i(\eta_{\text{ext}}, c_k, a_{ki}, \Gamma_i) = \Gamma_i + h(\eta_{\text{ext}}) + \sum_{k=1}^M a_{ki} r_k. \quad (9)$$

The point-source channel consists of a sum of M Gaussian functions r_k , whose amplitudes a_{ki} can vary between epochs, unlike their centers c_k , which are fixed for all epochs. The extended source channel consists of a grid of pixels $h(\eta_{\text{ext}})$ common to all epochs. Each pixel contained in η_{ext} is a free parameter. Γ_i accounts for imperfect sky subtraction by applying a spatially constant background correction.

Similar to Equation (4), the data fidelity term reads

$$\mathcal{L}(d | \eta_{\text{ext}}, a_{ki}, c_k, \Gamma_i) = \frac{1}{2} \sum_{i=1}^{n_{\text{epoch}}} [d_i - D S_i f_i]^\top C^{-1} [d_i - D S_i f_i], \quad (10)$$

where we have defined $\mathbf{S}_i$ as the convolution by s_i , the narrow PSF kernel corresponding to the i th epoch. Similar to the PSF reconstruction case, we apply a sparsity prior to regularize the extended channel contribution and the point-source channel to avoid flux leakage from the extended channel to the point-source channel. The deconvolution minimization problem can then be written as:

$$\begin{aligned} \arg \min_{\eta_{\text{ext}}, a_k, c_k, \Gamma_i} & \frac{1}{2} \sum_{i=1}^{n_{\text{epoch}}} [\mathbf{d}_i - \mathbf{D} \mathbf{S}_i \mathbf{f}_i]^T \mathbf{C}^{-1} [\mathbf{d}_i - \mathbf{D} \mathbf{S}_i \mathbf{f}_i] \\ & + \lambda \|\mathbf{W} \odot \Phi^T \mathbf{h}(\eta_{\text{ext}})\|_1 \\ & + \lambda_{\text{pts}} \left\| \mathbf{W} \odot \Phi^T \sum_{k=1}^M \langle a_k \rangle \mathbf{r}_k \right\|_1, \end{aligned} \quad (11)$$

where $\langle a_k \rangle$ is the mean amplitude over all epochs of the k th point source. If the data are sufficiently informative to lift the degeneracy between the point source and the background, λ_{pts} can be set to 0 to avoid the unnecessary computation of the second regularization term. This scenario occurs, for instance, when the point sources fade and are no longer detectable, as demonstrated in our example in Section 4.1. However, in situations where there is significant blending between the point source and the extended source and this degeneracy remains for all epochs, it becomes necessary to include this correction term. λ_{pts} can then be estimated iteratively, as indicated in Appendix C.

2.3. Noise Propagation

The weight matrix $\mathbf{W}$ in Equation (11) plays a central role in controlling the regularization strength, as it reweights the starlet coefficients according to their corresponding noise levels in the direct space. In other words, the coefficients of $\mathbf{W}$ are derived from the noise contained in the original images once propagated to the starlet coefficients of $\mathbf{h}$ (or $\mathbf{b}$ for the PSF formulation). This implies applying the transposed convolution operators to several noise realizations, before empirically evaluating the variance of the starlet coefficients (Joseph et al. 2019; Galan et al. 2022).

More specifically, a noise realization $\delta \mathbf{d}$ according to an upsampled version of the noise covariance matrix $\mathbf{C}$, denoted as $\mathbf{C}^\dagger$, is propagated to the grid of pixels as $\mathbf{S}_1^T (\mathbf{C}^\dagger)^{-1} \delta \mathbf{d}$, and as $\Phi^T \mathbf{S}_1^T (\mathbf{C}^\dagger)^{-1} \delta \mathbf{d}$ in the starlet space. In practice, we use a Monte Carlo approach to generate several realizations of $\delta \mathbf{d}$ and calculate the standard deviations of the starlet coefficients within each starlet scale. These correspond to the coefficients of the $\mathbf{W}$ matrix. Note that for multi-epoch deconvolution (or multistar PSF reconstruction), the noise realization $\delta \mathbf{d}$ can be drawn from the *effective* noise map, which corresponds to the noise map of the coadded data. The regularization strength hyperparameter λ is then expressed in units of noise standard deviation for the combined data set. For an illustration of this procedure, we refer the reader to Appendix B.

2.4. JAX Implementation

STARRED proposes a JAX implementation for the PSF reconstruction (Equation (7)) and deconvolution (Equation (11)) optimization problems. We use JAX automatic differentiation to compute the gradient of the loss function and accelerate the convergence by using a gradient-informed optimizer, such as *adabelief* or *l-bfgs-b* from the OPTAX and JAXOPT library

(DeepMind et al. 2020; Blondel et al. 2021). STARRED supports hardware acceleration on a GPU. The deconvolution runtime on a single consumer-grade GPU (NVIDIA RTX 3060) as a function of the input vector size is shown in Figure 2.

In the context of the large volume of data produced by LSST, STARRED will be capable of deconvolving and extracting the light curve for the nearly 1000 epochs collected during the 10 yr survey in approximately 10 s, utilizing just a single GPU (assuming 32×32 pixels input cutouts, deconvolved to 64×64 images). These 10 s include a compilation time of about 2–3 s, which is necessary only for the first run and can be avoided in any following deconvolution with the same input size. In this example, the runtime is about 30 times slower on a CPU.

Unlike deconvolution, PSF reconstruction cannot be performed for all the data at once and needs to be performed sequentially, as the PSF is expected to vary from epoch to epoch. However, this is easily parallelizable. Reconstructing the PSF for $6 \times 32 \times 32$ star cutouts takes approximately 2.5 s on a GPU. If the method is deployed on a large scale, the processing speed may not be sufficient for handling millions of cutouts across thousands of epochs in the LSST survey. The main bottleneck in this process is not so much the optimization runtime, which fully leverages GPU capabilities, but rather the input/output (I/O) and preprocessing times, which are CPU-dependent. However, the I/O runtime and the CPU/GPU memory transfer could be largely optimized on a high-performance computing cluster. Moreover, if computational speed remains a concern, a feasible alternative would be to use the precalculated PSF models that are provided as part of the LSST image data products, reserving the PSF reconstruction method solely for the highest-priority light-curve extractions. However, this may come with a loss in photometric accuracy, if the PIFF PSF modeling approach that is planned for the LSST image processing pipeline follows that of the Dark Energy Survey (Jarvis et al. 2021). Further investigation is needed, but this might just be a necessary tradeoff for certain science cases.

In summary, STARRED is already scalable for extracting the photometry of millions of transients found in the LSST survey if an accurate PSF model for each epoch is provided. Using the code to reconstruct the PSF for all these cutouts is likely achievable, but may still require some optimization.

3. PSF Reconstruction Performance

3.1. Metric Definitions

To evaluate the performance of the PSF reconstruction algorithm, we define four metrics, quantifying the astrometric precision, the photometric accuracy and precision, and the fidelity of the PSF reconstruction.

We define the astrometric precision as

$$P_{\text{astro}} = \frac{1}{N} \sum_{k=1}^N \|\mathbf{c}_k - \hat{\mathbf{c}}_k - \mathbf{b}\|_2, \quad (12)$$

where $\hat{\mathbf{c}}_k$ is the estimated astrometric position of the k th star and $\mathbf{c}_k$ is its input position in the simulation. $\mathbf{b}$ is the astrometric bias, defined as

$$\mathbf{b} = \frac{1}{N} \sum_{k=1}^N (\mathbf{c}_k - \hat{\mathbf{c}}_k). \quad (13)$$

We focus mostly on the astrometric precision, since a subpixel bias can be introduced during the forward simulation process while interpolating the data on a pixel grid.

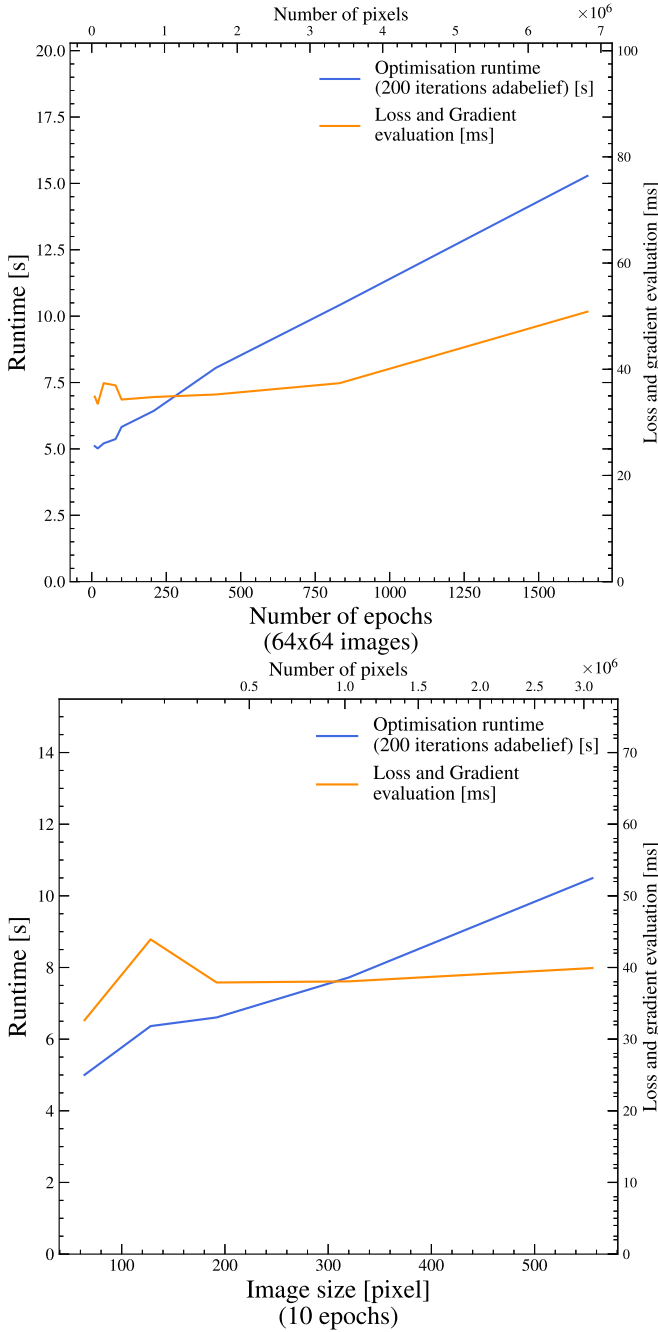


Figure 2. Upper panel: STARRED’s deconvolution runtime on an NVIDIA RTX 3060 GPU for the deconvolution of 32×32 input images (64×64 deconvolved images) as a function of the number of epochs. The full optimization runtime for 200 iterations of adabelief is shown in blue (in seconds), whereas the time needed for a single evaluation of the loss and its gradient is shown in orange (in milliseconds). Lower panel: STARRED deconvolution runtime for 10 epochs as a function of the output image size. These plots can be reproduced from this notebook [↗](#).

The relative photometric accuracy is defined as

$$A_{\text{photo}} = \frac{1}{N} \sum_{k=1}^N \frac{f_k - \hat{f}_k}{\hat{f}_k}, \quad (14)$$

while the relative photometric precision reads

$$P_{\text{photo}} = \frac{1}{N} \sum_{k=1}^N \left| \frac{f_k - \hat{f}_k}{\hat{f}_k} - A_{\text{photo}} \right|. \quad (15)$$

We also define a metric to quantify the fidelity of the PSF reconstruction, i.e., the mean absolute error (MAE) of the PSF:

$$\text{MAE}_{\text{PSF}} = \|\mathbf{PSF} - \widehat{\mathbf{PSF}}\|_1. \quad (16)$$

3.2. Results

In this first test, we simulated six high-SNR star cutouts as observed by the HST in the F814W filter. These mock observations were generated from the TinyTim PSF (Krist et al. 2011), subsampled by a factor of 10. We chose a random subpixel shift to mimic different star positions relative to the pixel grid and we then interpolated the 10-times subsampled TinyTim PSF before downsampling the simulated images by a factor of 5, to obtain a final pixel size of $0''.04$. Finally, we added a background and Poisson noise to the image and adjusted the amplitude of the stars to obtain an SNR of 800,000 for each star.

We ran three different PSF reconstruction algorithms, namely PHOTUTILS (Bradley et al. 2023), PSFR,⁷ and STARRED, on this simulated data set and compared the performance in terms of the fidelity of the reconstructed PSF, MAE_{PSF} , against the true original TinyTim PSF at the same resolution. In all three cases, we performed the reconstruction on a three-times subsampled pixel grid, corresponding to $0''.013$ per pixel. The results are shown in Figure 3.

In this high-SNR regime, STARRED accurately reconstructs the original PSF, especially in the center, where the natural dithering introduced by small and random subpixel shifts allows us to retrieve the highest frequencies. Although those frequencies are beyond the Nyquist frequency and are therefore not sampled for a single star, the random dithering between the images allows us to overcome this limit.

In terms of PSF fidelity, we obtain an MAE_{PSF} of 3.75×10^{-6} , 1.25×10^{-5} , and 1.70×10^{-5} for STARRED, PSFR, and PHOTUTILS respectively. This corresponds to an improvement factor of 3.3 and 4.5 in comparison to PSFR and PHOTUTILS.

We repeat this experiment for a range of SNRs to evaluate the performance of these methods in different SNR regimes. We used the four metrics defined in Section 3.1 to quantify the astrometric precision, the photometric precision and accuracy, and the fidelity of the PSF reconstruction. The results of those tests are summarized in Figure 4.

In terms of astrometric precision, STARRED outperforms PSFR and PHOTUTILS by almost an order of magnitude and in all SNR regimes. Notably, STARRED achieves this outcome by utilizing the random dithering among the six stars to reconstruct a subsampled version of the PSF, while the sparsity prior helps interpolating on a finer pixel grid. This leads to tight constraints on the positions of the stars. In this idealized scenario, where the six stars have an identical PSF and differ only in brightness, an astrometric precision better than a hundredth of an (original) pixel is attained for $\text{SNR} \geq 10,000$ and subsampling factor ≥ 2 . In real observations, we expect the precision to be degraded due to PSF field distortion and the spectral dependence of the PSF, as discussed in Section 5.

The third panel of Figure 4 shows the importance of reconstructing a supersampled version of the PSF to obtain unbiased photometric measurements. For $\text{SNR} \geq 1000$ and a supersampling factor of 3, the photometric accuracy is below

⁷ <https://github.com/sibirer/psfr>. This codes utilizes the features of Birrer et al. (2021).

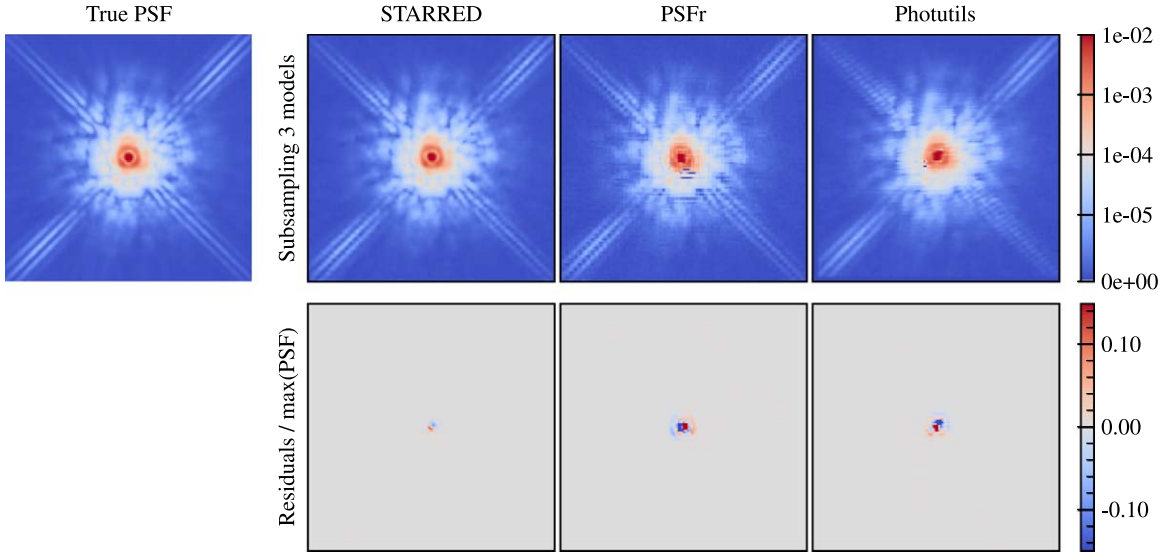


Figure 3. PSF reconstruction of the TinyTim PSF from six simulated stars with STARRED, PSFR, and PHOTUTILS (top row). The PSF is reconstructed with a pixel size three times smaller than that of the original data. The bottom row shows the difference between the reconstructed PSFs and the ground truth. This figure can be reproduced from this notebook [🔗](#).

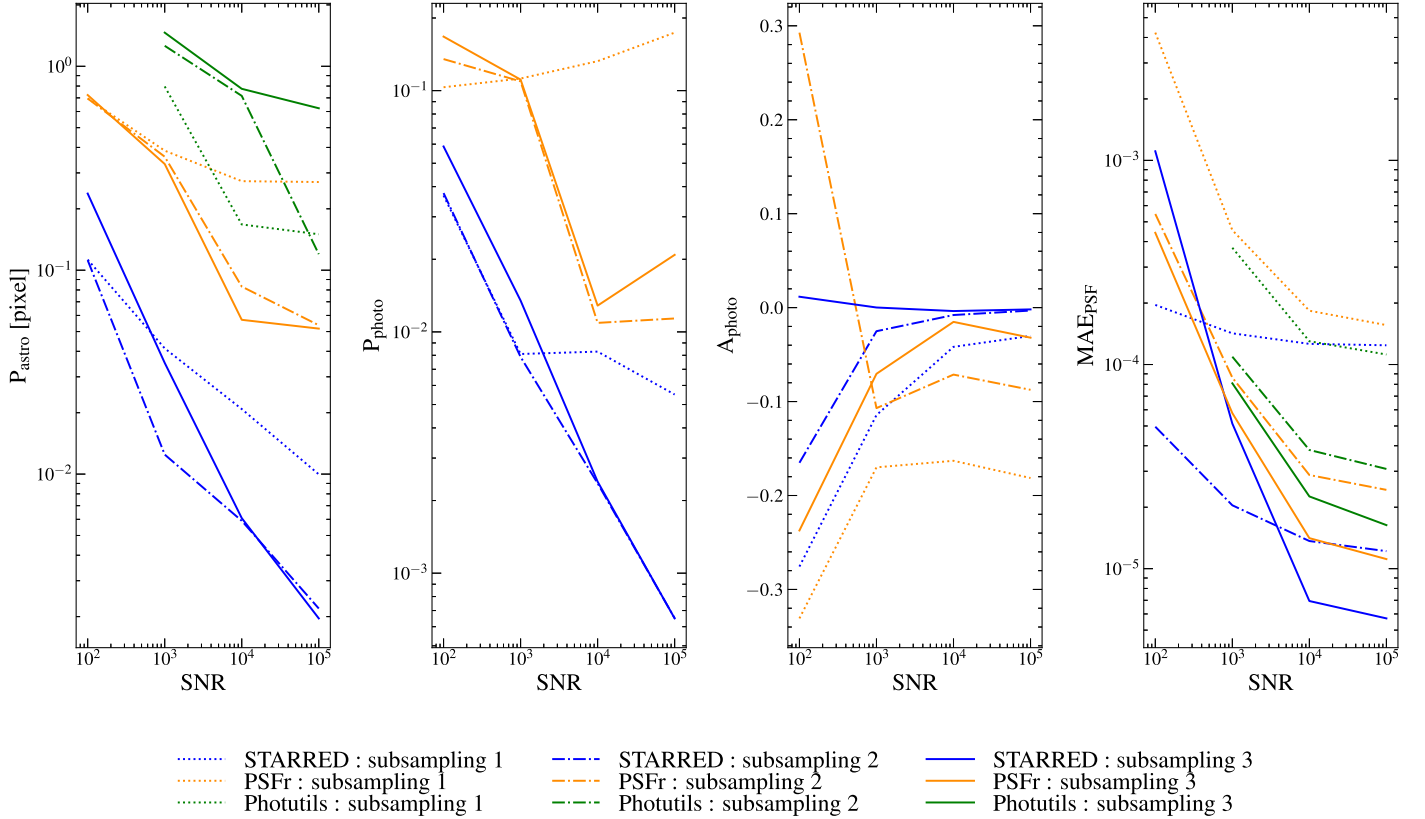


Figure 4. Astrometric precision (first panel), photometric precision (second panel), photometric accuracy (third panel), and PSF reconstruction fidelity (fourth panel) as a function of the SNR of the stars used to reconstruct the PSF. The dotted, dashed-dotted, and solid lines, respectively, indicate that the PSF is reconstructed at the same, at twice, or at three times the original pixel size. STARRED's performance (in blue) is compared to that of other empirical PSF reconstruction codes, such as PSFR (in orange) and PHOTUTILS (in green). The PHOTUTILS algorithm failed to converge at an SNR below 100 and the reconstructed PSF models were not sufficiently accurate to extract meaningful PSF photometry, which is why these results are not included in this figure.

0.3%, which corresponds to a few millimagnitudes. When the PSF is reconstructed on the same pixel grid as the original data, the photometry is underestimated by 3%–11% (for SNRs ranging from 1000 to 100,000). At high SNR, the PSF can be constrained to a finer grid of pixels, leading to a significant gain

in terms of astrometric precision, photometric precision, and photometric accuracy. However, at low SNR, most algorithms fail at reconstructing the PSF if the supersampling factor is too high. This results in a degradation of the MBE_{PSF} , as can be seen in the right panel of Figure 4.

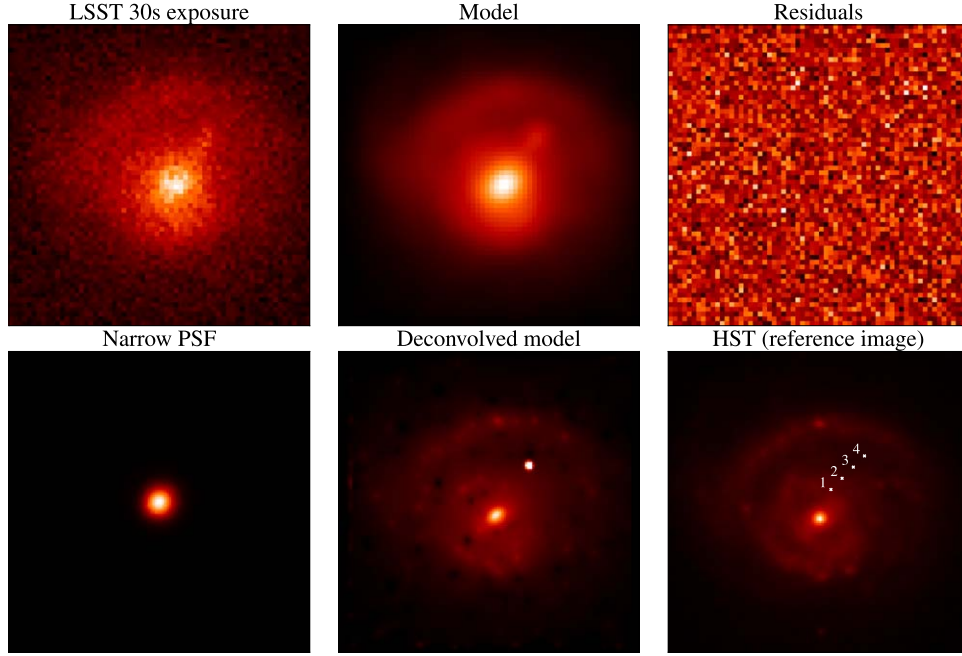


Figure 5. Multi-epoch deconvolution of a galaxy hosting an SN Ia. From top to bottom and from left to right, the different panels show a single-epoch image, the model produced by STARRED at the original resolution, the residuals for one epoch, the reconstructed narrow PSF s , the deconvolved model, and the reference HST image of the host galaxy. In the lower right panel, we indicate the four different locations chosen for the SN Ia explosion. This example corresponds to the fourth location, i.e., $2''$ from the center of the host galaxy. Note that the deconvolved model of the extended channel benefits from the SNR of the whole data set.

4. Deconvolution Performance

4.1. Example 1: Scene Modeling of Supernova Light Curves

Time-series simulation. In this first test example, we generate mock LSST observations of an SN Ia exploding in a spiral galaxy. The host galaxy is taken from the HST-COSMOS survey, which we degrade to ground-based resolution by convolving it with a set of empirical PSFs, directly modeled from real monitoring data taken at the 2.2 m MPIA telescope in La Silla, Chile (Millon et al. 2020a). The seeing ranges from $0''.61$ to $1''.47$, with a median at $0''.92$ (FWHM). The mock observations are generated using the image simulation module of *lenstronomy* (Birrer et al. 2021), assuming an LSST zero-point of 28.16 in the r band and a sky brightness of 21.2 mag per square arcsecond. We generate 50 epochs of observations at a cadence of 10 days. Each epoch consists of two 30 s visit images separated by 30 minutes (and assumed to have been taken in the same filter) and we introduce dithering randomly distributed around the center of the image and with a standard deviation of $0''.3$. For each epoch, we also generate six star cutouts to reconstruct the PSF. The PSF star’s R -band magnitude is normally distributed around 16, with a scatter of 0.3 mag.

After the 30th epoch, we inject a varying point source in the galaxy to simulate an SN Ia explosion. The light curves are interpolated from the *piscola* (Müller-Bravo et al. 2022) standard SN Ia template and scaled to reach a peak r -band magnitude of 19. We test four different locations of the SN Ia in the galaxy, marked as white dots on the bottom right panel of Figure 5, to investigate the effect of the local surface brightness on the deblending performance.

Light-curve extraction. First, we reconstruct the narrow PSF $s(\eta_{\text{PSF}})$ at each epoch by fitting the six star cutouts. Then we process all 100 images of the SN Ia host galaxy simultaneously to reconstruct a high-resolution image of the host galaxy (on a

grid of pixels subsampled by a factor of 2 compared to the original data) and to measure the PSF photometry at each epoch. In this example, there is no degeneracy between the point-source channel and the extended channel, since we process simultaneously the image before the SN Ia explosion. Therefore, λ_{pts} is set to 0. We also set $\lambda = 1$, meaning that starlet coefficients 1σ above the noise level are reconstructed in the high-resolution image. This reconstructed high-resolution image is shown in Figure 5 and the extracted PSF photometry is presented in Figure 6. For comparison, we also extract the light curve with the difference imaging analysis (DIA) presented in Dux et al. (2024), based on the methodology outlined in Alard & Lupton (1998). Difference imaging relies on the subtraction of the flux of all images from a reference epoch. To achieve perfect subtraction, the resolutions of the images need to coincide, which is achieved by blurring the best seeing image for comparison with every other image. With this method, the galaxy is perfectly subtracted, leaving only a single PSF corresponding to the flux excess of the SN Ia. The photometry can then be extracted with a simple aperture sum.

Results. We use similar metrics to those defined in Section 3 to evaluate the precision and accuracy of the recovered point-source photometry of the supernova. We compute the mean precision and the mean accuracy by comparing, after the supernova explosion, the measured flux to the true flux used in the simulation. We also measure the mean astrometric precision P_{astro} by comparing the recovered offset of the image and the true dithering shift input in the simulation. Those metrics are summarized in Table 1.

In this example, STARRED recovers the position of the supernova to a precision of about 0.08 pixels, corresponding to $0''.017$. In terms of photometric precision, STARRED outperforms the DIA method by a factor of 5–10, depending on the SN Ia location. The absolute photometry is also less biased, with a mean accuracy ranging from 1 to 37 mmag. The greatest

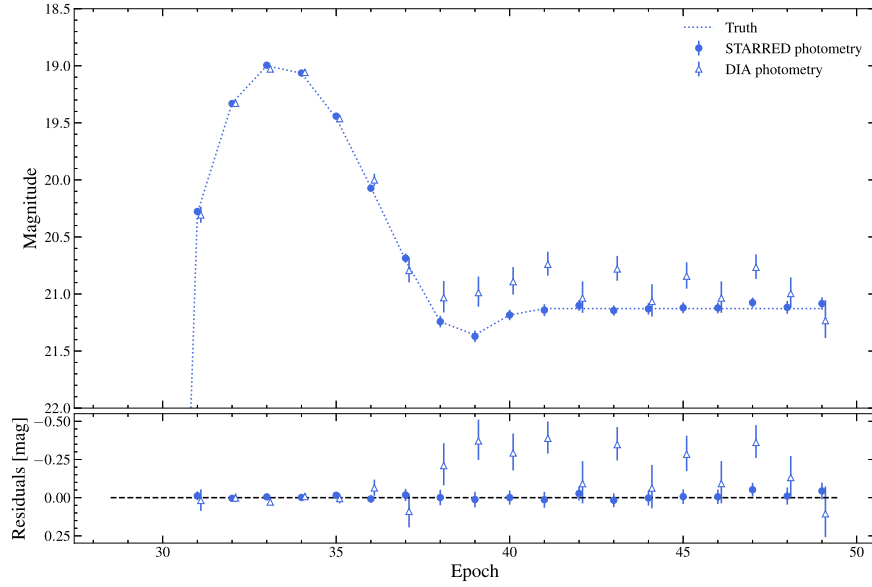


Figure 6. SN Ia light curve extracted with STARRED using the deconvolved model shown in Figure 5 and with DIA + aperture photometry. The bottom panel displays the residuals compared to the true light-curve input in the simulation. For the sake of clarity, the 30 observation epochs prior to the SN Ia explosion are not shown in this plot.

Table 1

Photometric Precision, Photometric Accuracy, and Astrometric Precision Recovered by Both STARRED and Difference Imaging Photometry for Four Different Locations of the SN Ia in the Host Galaxy

Location	SB (mag·arcsec ⁻²)	P_{starred} (mag)	P_{DIA} (mag)	A_{starred} (mag)	A_{DIA} (mag)	P_{astro} (pixel)
1	21.931	0.029	0.145	−0.037	−0.111	0.083
2	22.303	0.024	0.141	−0.028	−0.125	0.079
3	22.639	0.013	0.142	−0.008	−0.129	0.082
4	22.808	0.013	0.144	0.001	−0.129	0.081

Note. The light curves are derived from 100 simulated LSST observations of 30 s of exposure each. The second column indicates the surface brightness of the host galaxy at the location of the SN Ia.

bias occurs when the supernova is closest to the bulge of the lensing galaxy. This level of accuracy is considerably better than that of the DIA method, which typically shows a mean accuracy of 0.11–0.13 mag.

4.2. Example 2: Lensed Quasar Light Curves

For most lensed quasars, the main factor limiting photometric precision in their light curves is not the photon noise but the deblending errors, most often leading to systematics. Lensed quasars are indeed highly blended objects, since their image separation rarely exceeds 2'', and the light of the lens galaxy and from the gravitational arcs (hereafter, “the extended components”) is mixed with the quasar images. Neglecting the blending between the extended components and point sources in lens systems leads to increased scatter in the light curves, because these elements are intermixed in a manner influenced by the time-varying (and time-correlated) PSF. The issue is particularly pronounced in systems with small separations between images, where this mixing effect becomes more significant. It can be mitigated, however, by careful modeling of the PSF at each epoch and accurate modeling of the extended components, as we demonstrate below.

Extracting the light curves of lensed quasars is more difficult than in the case of supernovae, because the point source does not disappear after a few months. Instead, the point source

remains fully degenerate with the background, which makes measuring the magnitude of the quasar extremely challenging. In this example, we test the ability of STARRED to lift this degeneracy and obtain precise and accurate light curves of the multiple images of lensed quasars.

Time-series simulation. We use LENSTRONOMY to generate mock images of gravitationally lensed quasars, following a procedure akin to the one described in Section 4.1, i.e., assuming the same observational and instrumental parameters, including the same empirical PSF as a function of time. We model the lens galaxy as a Sérsic profile with Sérsic index $n_{\text{Sérsic}} = 3$ for the light distribution and as a singular isothermal ellipsoid for the mass distribution. The source galaxy is modeled as an exponential profile (i.e., $n_{\text{Sérsic}} = 1$) and a quasar, represented as a point source. We choose a close alignment between the source and the lens to ensure the formation of four multiple images of the quasar, in an Einstein cross-configuration. We impose different Einstein radii for the lens, ranging from $\theta_E = 0''.5$ to $\theta_E = 1''.5$, to test how STARRED performs in deblending highly compact lens systems. This range is particularly relevant for galaxy-scale lenses and useful for time-delay cosmography, as more compact systems may be difficult to detect.

For the variability of the quasar, we take the real light curve of the lensed quasar RXJ1131–1231, as observed by the Euler Swiss Telescope in the 2013 season (Millon et al. 2020b). The

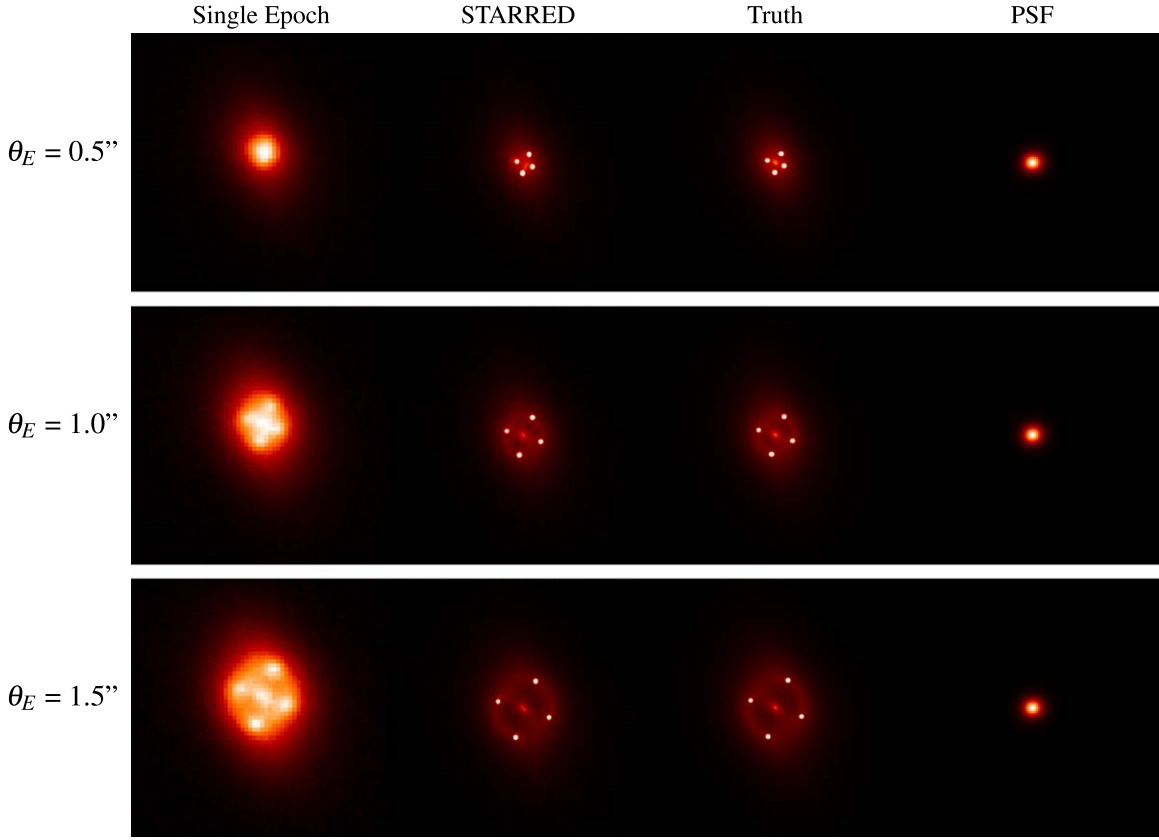


Figure 7. Multi-epoch deconvolution of a quadruply imaged quasar for different Einstein radii θ_E . From left to right, the panels show single-epoch simulated LSST images, the deconvolved models reconstructed from all 100 images, and the truth images at the same resolution as the deconvolved image and the PSF.

light curves are interpolated and rescaled to a mean magnitude of 21.5, before lensing magnification. Since we only intend to test the light-curve extraction, we assume that there is no time delay between images, so all images vary simultaneously. This is actually the worst-case scenario, as it maximizes possible flux leakage between point sources, making the light curves maximally degenerate. The cadence of observation is set to 10 days.

Results. Using the same reconstructed PSF as in the previous example, we run the deconvolution on all 100 images simultaneously, fitting for both the position and photometry of the multiple images of the quasar. The extended channel is used to reconstruct a model of the light profile of the lens galaxy and of the Einstein ring, at three times the sampling of the original image. The best-fit deconvolved models are shown in Figure 7 and the extracted light curves for each of the four multiple images are shown in Figure 8. For comparison, we also extract the photometry with PHOTUTILS in an aperture of 8 pixels ($1''.68$ in diameter) for each images. The background is estimated in an annulus of 2 pixels around the aperture. Table 2 provides a summary of the mean photometric precision and accuracy achieved with both STARRED and aperture photometry using PHOTUTILS.

The recovered mean photometric precision over all 50 observation epochs ranges from 21 to 133 mmag, depending on the image brightness and on the compactness of the lensing configuration. This exhibits a precision approximately two to five times higher compared to the photometry obtained with aperture photometry. In terms of accuracy, the absolute brightness of the images is recovered to less than 0.1 mag in most lensing configurations, except for the most compact one

($\theta_E = 0''.5$), where the photometric bias reaches 0.19 mag for image B. This is particularly advantageous, as we intentionally selected a lens system featuring a bright Einstein ring and a bright lens galaxy to assess the deblending capability of STARRED. Even if the photometry of one multiple image can be slightly over- or underestimated due to residual deblending errors, the peak in the light curves around epoch 40 is still clearly identified and this does not compromise our ability to measure a time delay. However, for other science cases requiring a measurement of the flux ratio, this systematic error should be carefully quantified and integrated into the overall uncertainty calculations.

Similar to our previous example, the position of the multiple images is also recovered with a precision of about a tenth of pixel.

4.3. Example 3: Photometry of High-redshift Globular Clusters with JWST

A generic application of STARRED is the photometry of point sources superposed on a complex luminous scene. This is well illustrated by the photometry of compact sources associated to the “Sparkler” galaxy (Mowla et al. 2022) from a single epoch of observation. The “Sparkler” is a galaxy at redshift $z = 1.378$ strongly lensed by the $z = 0.39$ galaxy cluster SMACS J0723.3-7327. Most of these compact sources (“the sparkles”) are unresolved, even with JWST, and five of them were identified as globular clusters by Mowla et al. (2022), owing to their red SED compatible with the color of old stellar systems. However, Mowla et al. (2022) used aperture photometry to measure the SED of these sources, and most of them appear to

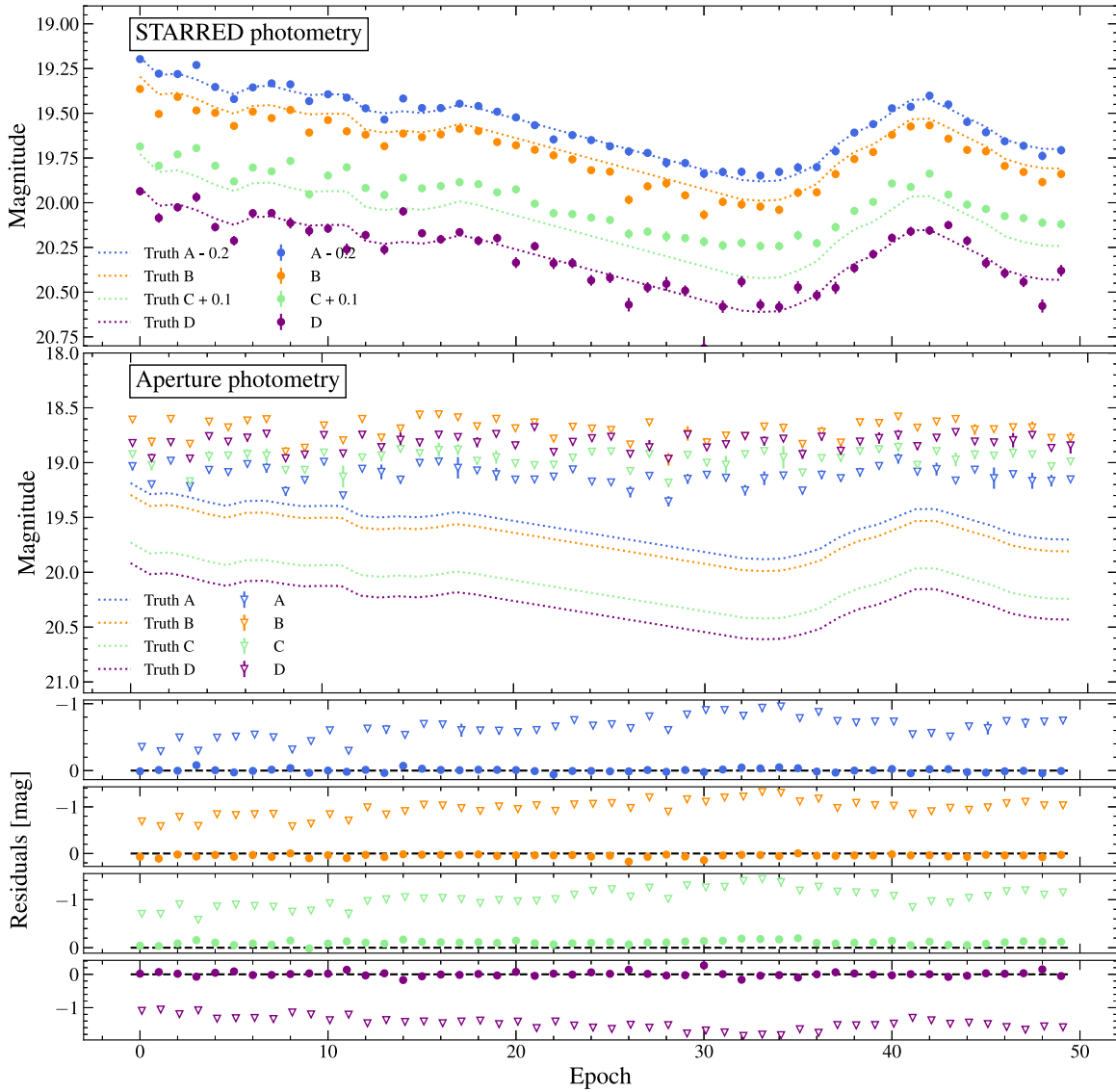


Figure 8. Lensed quasar light curves extracted with STARRED using the deconvolved model shown in the second row of Figure 7 ($\theta_E = 1''$). Aperture photometry obtained after annulus background subtraction is shown with triangles. The bottom panels show the residuals from each lensed image.

be contaminated by the extended galaxy emission. To demonstrate the deblending capability of STARRED on real JWST observations, we apply our two-channel deconvolution method, modeling the compact sources in the point-source channel and the contaminating light from the galaxy in the extended source channel. We use the reduced NIRCcam images of SMACS J0723.3-7327 in the F090W, F150W, F200W, F277W, F356W, and F444W bands provided by Mowla et al. (2022). These images have been reduced with a combination of a modified JWST pipeline and the Grizly software (Brammer & Matharu 2021), and they have a pixel scale of $0''.04$ per pixel. All the details about the image processing are provided in Noirot et al. (2023).

First, we estimate the narrow PSF in each band using four bright stars in the vicinity of the Sparkler. We then use the reconstructed narrow PSFs to perform a deconvolution of the data, improving the sampling by a factor of 2. The reconstructed PSF in the F200W band can be seen in the last panel of Figure 11. The original and deconvolved images are shown in Figure 9, where the identified point sources are labeled from 1 to 18. In addition to the 12 sources measured in

Mowla et al. (2022), we include five additional highly blended sources located in the south of the gravitational arc and a faint one to the north of the arc (#18).

Results. We confirm the point-source nature of 13 out of the 18 compact objects modeled in the point-source channel. Source IDs 5, 6, 7, 11, and 12 are probably extended, as some flux residuals are visible in the extended channel near the point sources. From the deblended photometry, we show the SEDs of these objects in Figure 10, while a table collecting all the measurements is presented in Appendix A. The aperture photometry measured in Mowla et al. (2022) is shown with dotted lines in Figure 10. As our photometric measurements mostly agree for the unblended systems, we observe important differences for the source IDs 4, 5, 7, 8, 10, 11, and 12. The differences for the most blended systems—IDs 7, 11, and 12, located in the center of the galaxy—reach a factor of 2–3 difference, highlighting the need for proper modeling of the background emission to obtain accurate photometry. Source IDs 5 and 6 are extended, which explains the difference between the PSF photometry performed by STARRED and the aperture photometry of Mowla et al. (2022). For the other point

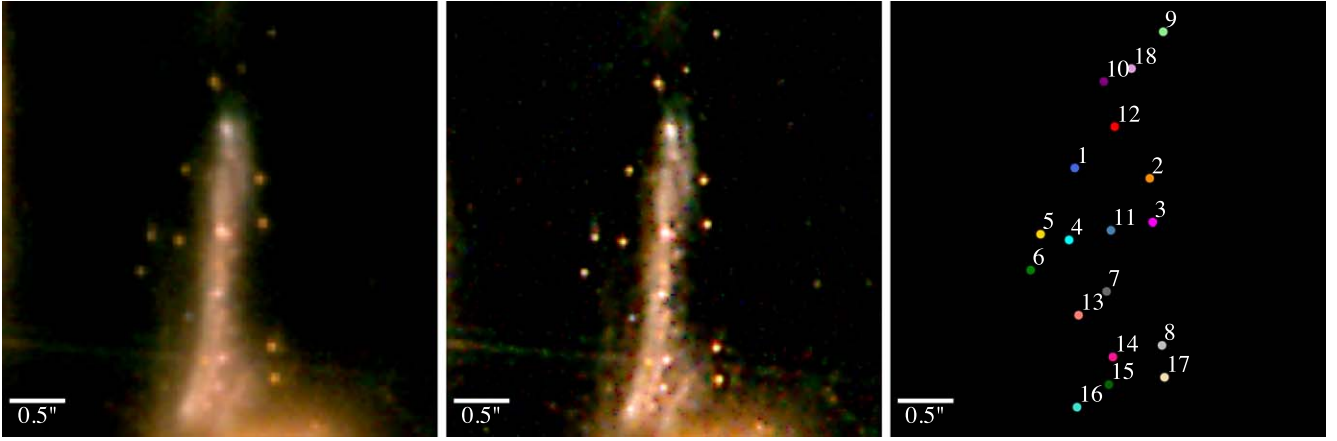


Figure 9. Left panel: JWST/Nircam color composite (R: F200W; G: F150W; B: F090W) image of the “Sparkler,” a strongly lensed galaxy by the galaxy cluster SMACS J0723.3-7327 (R.A. = 7:23:21.8, decl. = −73:27:18.7). The pixel size is $0''.04$ per pixel. Middle panel: STARRED deconvolved model. The pixel size is two times smaller than the original image, i.e., $0''.02$ per pixel. Right panel: location and ID number of the “sparkles,” modeled as point sources.

Table 2

Mean Photometric Precision and Accuracy Recovered by STARRED and Aperture Photometry (Including Annulus Background Subtraction) with PHOTUTILS for the Simulated LSST Images of a Quadrupty Imaged Lensed Quasar

θ_E (arcseconds)	Image	P_{starred} (mag)	P_{aperture} (mag)	A_{starred} (mag)	A_{aperture} (mag)	P_{astro} (pixel)
0.5	A	0.076	0.149	0.089	−1.526	0.134
	B	0.076	0.162	0.187	−1.771	0.174
	C	0.077	0.150	−0.002	−1.837	0.097
	D	0.133	0.166	0.009	−2.543	0.364
1.0	A	0.021	0.129	−0.002	−0.638	0.089
	B	0.024	0.136	0.050	−0.966	0.092
	C	0.031	0.149	−0.108	−1.037	0.085
	D	0.049	0.150	0.007	−1.472	0.078
1.5	A	0.023	0.149	0.028	−0.622	0.086
	B	0.024	0.121	0.102	−0.605	0.088
	C	0.026	0.150	−0.061	−0.965	0.076
	D	0.037	0.154	−0.049	−1.182	0.075

Note. The last column indicates the mean astrometric precision recovered by STARRED.

sources, we are able to deblend them from the extended emission for the following two reasons. First, the degeneracy between the extended background and the point sources is not perfect, as the shape of the PSF is known to high precision. Therefore, all the features in the PSF (including the diffraction spikes, the Airy rings, etc.) are used to constrain the amplitude of the point source. Second, the starlet prior is informing the extended source morphology “under” the point sources. PSF features wrongly attributed to the extended source channel are penalized by this prior. Some of the newly measured point sources (IDs 13–18) have similar SEDs to the globular clusters identified in Mowla et al. (2022; IDs 1, 2, 4, 8, and 10). Future NIRSpect integral field unit data will confirm (or not) our findings.

5. Discussion

For time-delay cosmography applications, achieving high photometric precision is essential, because it enables the clear identification of features that can be unequivocally matched

across the different light curves, facilitating accurate time-delay estimations. While photometric accuracy plays a secondary role for time-delay cosmography, it is critical in microlensing studies. The flux ratios are in fact rarely used for lens modeling, but are critical observables when studying microlensing caused by a star in the lens galaxy or millilensing by subhalos intervening along the line of sight (Schechter & Wambsganss 2002). Moreover, imperfect deblending can create misleading contrasts between the time-varying point sources and the static extended sources, leading to false microlensing signals in the differential light curve (see Sluse et al. 2006; Millon et al. 2020a for discussions on this issue).

Difference imaging and aperture photometry might be valid competing approaches for light-curve extraction of supernovae, as these transients eventually disappear, offering the opportunity to measure accurately the surface brightness at their location. However, difference imaging is certainly not a solution for small-separation lensed quasars and supernovae, as it requires downgrading the resolution of all but the best seeing image. In addition, difference imaging still requires the deblending of the residuals (i.e., the photometric variations) in each single-epoch residual frame. This can be done for a handful of bright and well-separated lenses (e.g., Sorgenfrei et al. 2024), but the difference imaging method is unsuitable for more compact systems. By contrast, we demonstrate that STARRED provides an accurate estimate of point-source magnitudes by modeling the extended components of the lens, lifting the remaining degeneracy by applying a morphology prior on the extended sources. This feature can also be useful for single-epoch applications, as we have shown in Section 4.3, where we obtained deblended photometry of globular clusters, separating the point-like components from the extended component of the host galaxy. Finally, STARRED considered the whole data set jointly, hence the total SNR rather than the single-epoch one. STARRED can therefore also be seen as way of combining images optimally, correcting for the PSF and naturally drizzling the resulting image.

This deblending capability is especially critical for compact lensed systems ($<1''$ image separation), which are expected to constitute most of the remaining unknown population, since most of the well-separated lensed quasars have already been found in static imaging surveys such as the Dark Energy Survey or Gaia (see, e.g., Figure 19 in Lemon et al. 2023). The

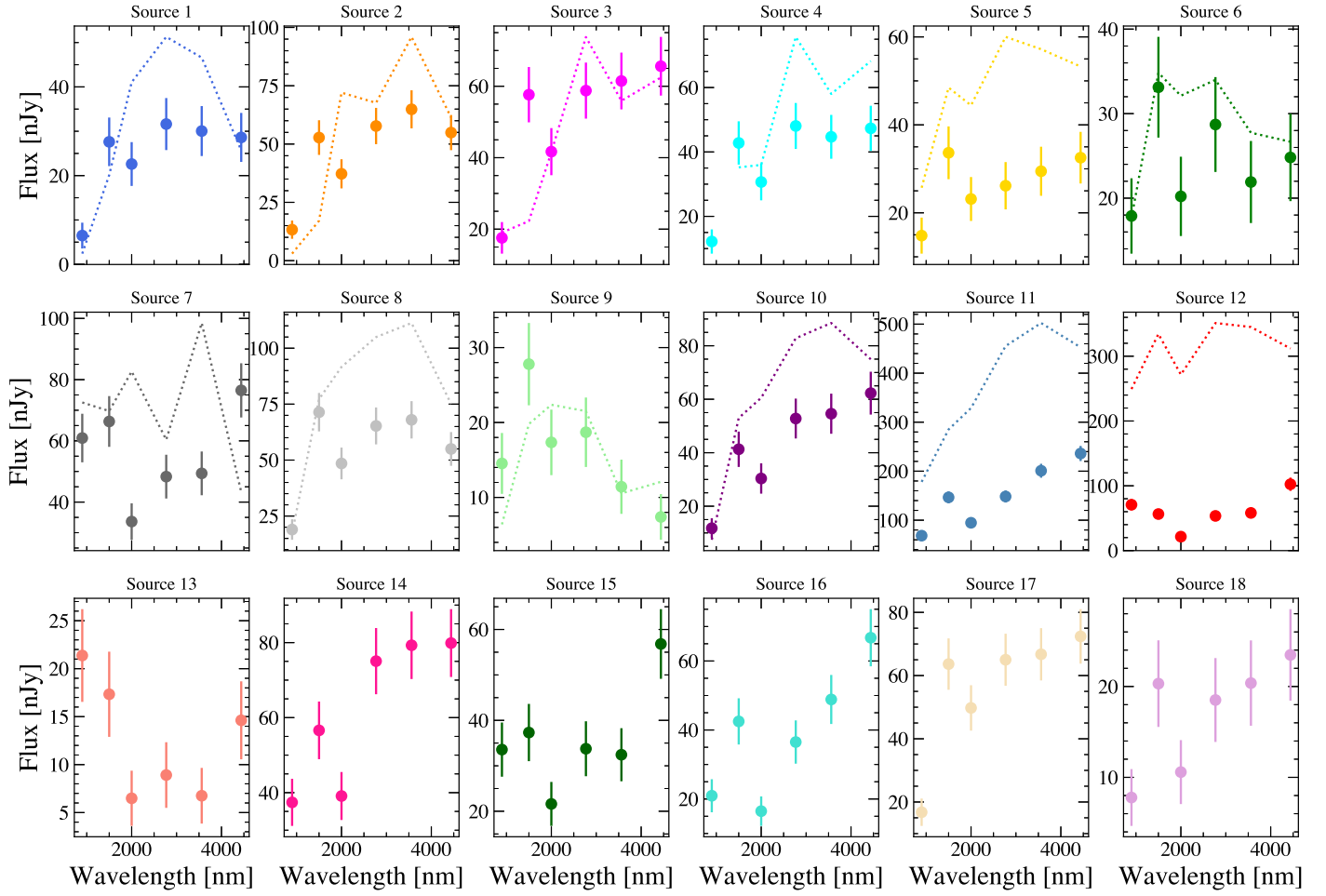


Figure 10. SEDs of the compact objects labeled in Figure 9 in the observer frame. The point-source photometry is measured from the deconvolved images. The dotted lines correspond to the SEDs measured in Mowla et al. (2022) using aperture photometry. Note that point sources #13–#18 were not measured previously. Some of them have SEDs compatible with those of globular clusters, such as the ones in Mowla et al. (2022). In addition, the aperture photometry of point sources exhibiting the closest alignment with the underlying arc is clearly biased compared to STARRED.

temporal dimension provided by the LSST survey leaves considerable hope of detecting these systems through their temporal variability, complementing previous lens searches (see, e.g., Taak & Treu 2023). Turning these systems into useful cosmological probes will require the resolution improvement capability of STARRED to reduce the scatter in the light curves and obtain robust time-delay estimates. Moreover, we demonstrate that STARRED is capable of recovering the point-source positions to a tenth of pixel. This is typically $0''.02$ for realistic LSST simulated images in average seeing conditions ($0''.92$ FWHM median seeing). This is an important result for many science applications and especially for time-delay cosmography, since most of the lensed quasars found in LSST will have limited space-based follow-up to determine the astrometry of the lensed image. Obtaining precise astrometry from the ground is already crucial for accurate lens modeling and to avoid bias on H_0 .

The choice of starlets for the regularization in STARRED is well suited for the representation of generic astronomical objects, but it presents limitations for representing Einstein rings. Other families of wavelets, such as curvelets (Starck et al. 2006), might be a better choice for representing lensed arcs, while STARRED is designed to maintain a generic approach to accommodate a wide range of applications. The implementation of alternative

regularization schemes for specific applications is left for future work.

Future improvements also include learning a prior on galaxy morphology using high-resolution imaging surveys like Euclid or Roman. On this matter, Adam et al. (2023) recently introduced a Bayesian framework that employs score-based likelihood characterization and diffusion models, allowing the retrieving of posterior samples of deconvolved galaxy light profiles from the data and using a prior learned from real observations. As STARRED is implemented in JAX and is autodifferentiable, it would be straightforward to combine it with machine-learning-based techniques. Promising methods also include hybrid approaches, such as Learnlets (Ramzi et al. 2023), which involves learning the dictionary for regularization from actual data. This method aims to merge the strengths of sparsity-based methods, such as exact reconstruction and strong generalization, with the rapid processing and flexible prior learning capabilities of neural networks.

In its current state, our method does not support multiband data nor does it account for PSF field distortion. Addressing the latter would require the implementation of an iterative correction of the PSF during the deconvolution process or, alternatively, inferring the full distortion field during the PSF reconstruction step. Finally, incorporating color information could significantly facilitate the separation of point sources

from extended background sources, as they often exhibit markedly different colors. This would require the combination of STARRED with existing sparsity-based deblending tools, such as MUSCADET (Joseph et al. 2016), which leverages galaxy morphology and color information to perform the deblending of extended sources.

6. Conclusion

In this work, we have presented a new PSF reconstruction, deconvolution, and light-curve extraction software based on the guiding principles of the MCS/Firedec algorithms. STARRED builds on the main concepts of its predecessors, deconvolving images into two distinct channels: one for point sources and the other for extended sources. However, STARRED features some major improvements compared to these earlier methods:

1. It introduces a new regularization based on starlets, a wavelet family well suited to representing astronomical objects.
2. STARRED is implemented in JAX and is completely autodifferentiable, ensuring a much faster convergence due to the utilization of gradient-informed optimizers.
3. STARRED is GPU-compatible, making the code 20–30 times faster than its predecessors. It also offers the possibility to process all observation epochs simultaneously, adequately modeling possible shifts and rotations between frames.
4. It introduces a new empirical PSF reconstruction technique, also based on sparse regularization. STARRED outperforms other empirical PSF reconstruction algorithms such as PHOTUTILS or PSFR in terms of the fidelity of the reconstructed PSF, photometric accuracy, and astrometric precision. This is due to the sparse regularization, which effectively denoises the reconstructed PSF. STARRED leverages the small subpixel shifts between stars relative to the pixel grid to reconstruct a subsampled version of the PSF. Obtaining a subsampled PSF is fundamental given the principle of the method, aiming at an improved resolution rather than an infinite one, which requires that the deconvolved model, and therefore the PSF, can be measured on a fine grid of pixels. This is especially true with poorly sampled space data (see, e.g., Symons et al. 2021). We also emphasize that the high-quality PSF reconstruction that STARRED offers has diverse science applications, including lens modeling (Galan et al. 2024) or shear measurements (Finner et al. 2023).

As implemented, the method has a wide range of applications, as it uses a generic PSF reconstruction algorithm and a robust two-channel deconvolution method. It is optimized for light-curve extraction of point-like objects, but of course also works with single-epoch observations or without any point sources in an image, still providing an optimal way of combining multi-epoch observations with variable PSFs. In this study, we limit ourselves to three main science cases using simulated LSST images and real JWST observations, arriving at the following findings:

1. For transient objects like SNe Ia, STARRED is capable of recovering unbiased photometry by accurately modeling the host galaxy surface brightness. Of note, as the data are

deconvolved jointly, the extended channel of the final image has the SNR of the whole data set, hence improving over the individual treatment of a single epoch by about the square root of the number of epochs.

2. Light curves of lensed images can be extracted with high photometric precision from LSST data, even for the most compact lensed quasars ($\theta_E < 1''$). This might open possible time-delay measurements for an entirely new population of compact lensed systems.
3. Point-source photometry can be extracted from single-epoch JWST observations with STARRED, despite the complex PSF of this telescope. Using the Sparkler galaxy as an example, we have demonstrated the utility of deblending point-like globular clusters from the extended host galaxy to obtain an unbiased SED.

Future enhancements of our method could involve correcting for changes in the PSF across the field of view. This correction would be especially crucial for adaptive optics observations, which suffer from significant PSF distortions throughout the field and chromatic dependence (Liaudat et al. 2023). Additionally, deep learning methods may be integrated to learn a more tailored prior on galaxy morphology, improving the generic starlet regularization currently in use. The wavelet basis could be optimized by using Learnlets (Ramzi et al. 2023), without losing any generality. By leveraging the abundance of high-resolution data from upcoming space-based surveys like Euclid, future hybrid techniques may provide the best of both worlds: they could learn meaningful priors on galaxy morphology while maintaining the precise reconstruction capabilities inherent to sparsity-based techniques.

Acknowledgments

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Data Availability

The JWST data presented in this article were obtained from the Mikulski Archive for Space Telescopes (MAST) at the Space Telescope Science Institute. The specific observations analyzed can be accessed via doi.org/10.17909/jgpd-kj32.

Software: STARRED (Michalewicz et al. 2023), LENSTRONOMY (Birrer et al. 2021), PHOTUTILS (Bradley et al. 2023), PSFR (Birrer et al. 2021), JAX (Bradbury et al. 2018), JAXOPT (Blondel et al. 2021), OPTAX (DeepMind et al. 2020), ASTROPY (Astropy Collaboration et al. 2022).

Appendix A Sparkler Photometry

The photometry of the compact sources around the Sparkler Galaxy is listed in Table 3.

Table 3
PSF Photometry in the Different JWST/Nircam Bands for 18 Compact Sources Identified around the Sparkler Galaxy

ID	R.A.	Decl.	F090W (nJy)	F150W (nJy)	F200W (nJy)	F277W (nJy)	F356W (nJy)	F444W (nJy)
1	07:23:21.8648	-73:27:18.2528	6.5 ± 2.9	27.6 ± 5.5	22.6 ± 4.9	31.6 ± 5.9	30.0 ± 5.6	28.6 ± 5.5
2	07:23:21.6948	-73:27:18.3440	13.3 ± 3.9	52.8 ± 7.4	37.2 ± 6.2	57.7 ± 7.8	64.9 ± 8.2	54.9 ± 7.5
3	07:23:21.6927	-73:27:18.7617	17.6 ± 4.4	57.6 ± 7.7	41.7 ± 6.6	58.8 ± 7.8	61.5 ± 7.9	65.6 ± 8.2
4	07:23:21.8770	-73:27:18.9312	12.2 ± 3.8	42.8 ± 6.7	30.7 ± 5.7	48.1 ± 7.1	44.7 ± 6.8	47.3 ± 7.0
5	07:23:21.9393	-73:27:18.8505	14.8 ± 4.1	33.6 ± 6.0	23.1 ± 5.0	26.2 ± 5.4	29.5 ± 5.6	32.5 ± 5.9
6	07:23:21.9557	-73:27:19.1906	17.9 ± 4.5	33.1 ± 6.0	20.2 ± 4.7	28.7 ± 5.6	21.9 ± 4.9	24.8 ± 5.2
7	07:23:21.7966	-73:27:19.3890	60.9 ± 7.9	66.3 ± 8.3	33.6 ± 6.0	48.3 ± 7.1	49.4 ± 7.1	76.5 ± 8.9
8	07:23:21.6719	-73:27:19.9046	19.0 ± 4.6	71.4 ± 8.6	48.5 ± 7.1	65.3 ± 8.2	68.0 ± 8.3	55.0 ± 7.5
9	07:23:21.6896	-73:27:16.9919	14.5 ± 4.1	27.8 ± 5.5	17.3 ± 4.4	18.7 ± 4.6	11.4 ± 3.6	7.4 ± 3.0
10	07:23:21.7977	-73:27:17.4499	11.8 ± 3.7	41.3 ± 6.6	30.4 ± 5.7	52.8 ± 7.5	54.6 ± 7.5	62.3 ± 8.0
11	07:23:21.7827	-73:27:18.8342	68.4 ± 8.4	146.6 ± 12.2	94.7 ± 9.8	148.4 ± 12.3	200.9 ± 14.2	236.1 ± 15.4
12	07:23:21.7790	-73:27:17.8600	70.7 ± 8.5	56.4 ± 7.7	21.5 ± 4.8	53.5 ± 7.5	58.1 ± 7.7	102.4 ± 10.2
13	07:23:21.8462	-73:27:19.5638	21.4 ± 4.8	17.3 ± 4.4	6.5 ± 2.9	8.9 ± 3.4	6.8 ± 2.9	14.6 ± 4.1
14	07:23:21.7762	-73:27:20.0096	37.4 ± 6.3	56.6 ± 7.7	39.1 ± 6.4	75.1 ± 8.8	79.3 ± 9.0	79.9 ± 9.0
15	07:23:21.7977	-73:27:20.2697	33.6 ± 6.0	37.3 ± 6.3	21.6 ± 4.8	33.8 ± 6.0	32.4 ± 5.8	56.8 ± 7.7
16	07:23:21.8571	-73:27:20.5067	20.9 ± 4.8	42.5 ± 6.7	16.5 ± 4.3	36.5 ± 6.3	48.9 ± 7.1	66.7 ± 8.3
17	07:23:21.6648	-73:27:20.2113	16.8 ± 4.3	63.6 ± 8.1	49.8 ± 7.2	65.0 ± 8.2	66.7 ± 8.3	72.4 ± 8.6
18	07:23:21.7412	-73:27:17.3260	7.8 ± 3.1	20.3 ± 4.8	10.6 ± 3.5	18.5 ± 4.6	20.4 ± 4.7	23.5 ± 5.0

Appendix B Noise Propagation in Starlet Space

As an illustration of the noise propagation procedure described in Section 2.3, we show in Figure 11 the estimated weight matrix W when reconstructing the JWST/Nircam

F200W PSF used in Section 4.3. The first seven panels depict the coefficients of the W matrix for each of the seven starlet scales. W is used to reweight the starlet coefficients before computing the ℓ_1 -norm.

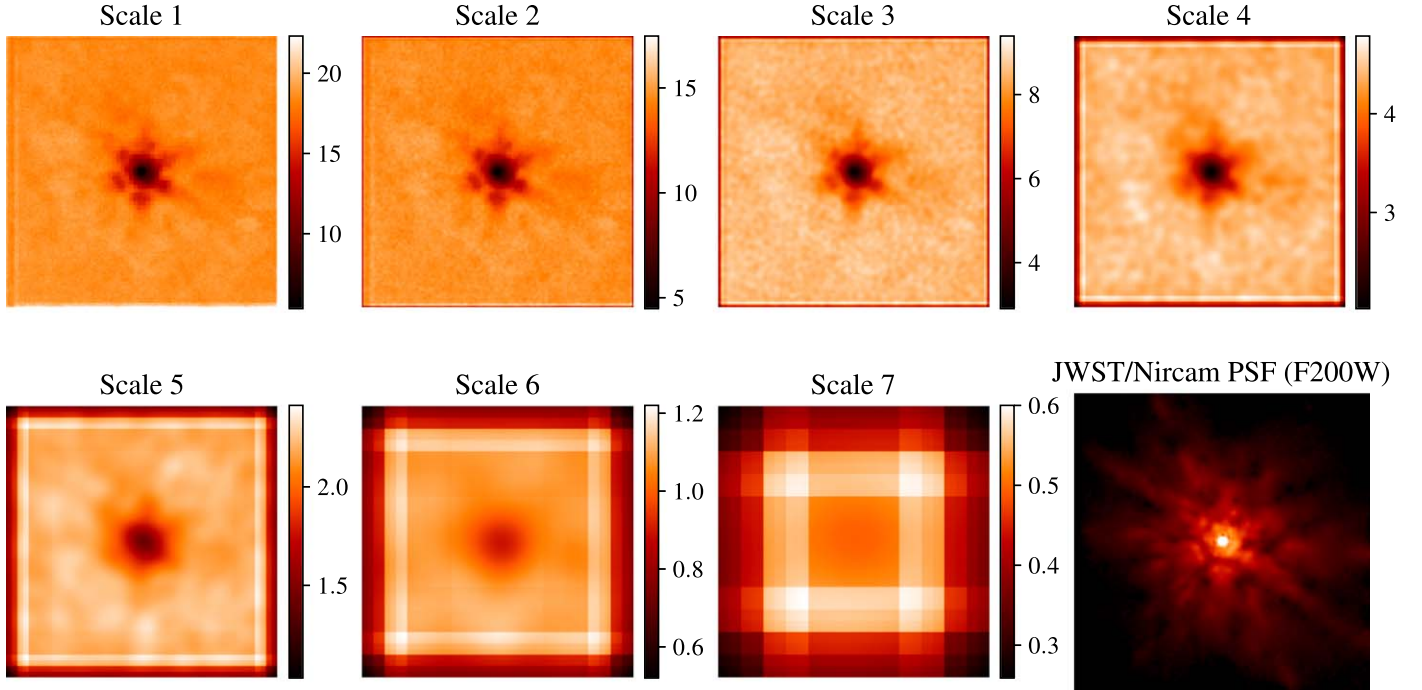


Figure 11. Weight matrix W used in the regularization term of Equation (6). It is computed in a Monte Carlo procedure by generating 5000 random noise realizations, propagating it into the starlet space, and measuring the variance of each starlet scale. In the last panel, we show our reconstructed model of the JWST/Nircam PSF in the F200W band at a sampling of $0''.013$ per pixel. The PSF has been reconstructed from four stars located near the Sparkler galaxy in the field of view of the galaxy cluster SMACS J0723.3-732. This figure can be reproduced from this notebook [🔗](#).

Appendix C

Regularization of the Point-source Channel

The MCS and Firedec algorithms did not incorporate regularization for the point-source channel, leading to the frequent formation of “holes” in the extended channel beneath the point sources. This was due to the lack of regularization of the point sources, resulting in them being overweighted in the loss function compared to the extended channel, which was subjected to the sparsity constraint. To address this issue, STARRED introduces a correction that compensates for the flux artificially absorbed by the point sources, while keeping a morphology prior on the extended sources.

The correction term λ_{pts} in Equation (11) can be estimated from the difference of the ℓ_1 -norm of the image with and without the point-source channel, normalized by the sum of the ℓ_1 -norm of the individual point sources. The correction factor to apply reads

$$\lambda_{\text{pts}} = \lambda \frac{\| \mathbf{W} \odot \Phi^T \mathbf{f} \|_1 - \lambda \sum_{k=1}^M \langle a_k \rangle \| \mathbf{W} \odot \Phi^T \mathbf{r}_k \|_1}{\sum_{k=1}^M \langle a_k \rangle \| \mathbf{W} \odot \Phi^T \mathbf{r}_k \|_1}, \quad (\text{C1})$$

where $\mathbf{f}$ is the average of all single-epoch improved-resolution images $\mathbf{f}_i$ and $\langle a_k \rangle$ is the mean amplitude of the k th point source. Note that in general

$$\sum_{k=1}^M \langle a_k \rangle \| \mathbf{W} \odot \Phi^T \mathbf{r}_k \|_1 \neq \left\| \mathbf{W} \odot \Phi^T \sum_{k=1}^M \langle a_k \rangle \mathbf{r}_k \right\|_1, \quad (\text{C2})$$

so the regularization term in Equation (11) does not reduce to $\lambda \| \mathbf{W} \odot \Phi^T \mathbf{f} \|_1$.

In practice, recomputing the ℓ_1 -norm of $\mathbf{f}$, $\mathbf{h}(\eta_{\text{ext}})$, and $\mathbf{r}_k$ at each iteration could lead to convergence problems and be computationally expensive. For that reason, we choose to initially set λ_{pts} to 0.5λ as a first approximation and iteratively refine it from the best-fit model in the previous iteration until convergence is reached.

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